DASINI RATIYALA

Software Engineering Undergraduate

0702089601 / 0757639179 | ratiyaladasini@gmail.com | Horagala, Padukka

linkedin.com/in/dasini-ratiyala-5630122a1 | github.com/dasi005



SUMMARY

Software Engineering undergraduate pursuing a BSc (Hons) from the University of Plymouth (UK) at NSBM Green University, with a Second Class Honours (Upper Division) standing. Hands-on experience building mobile, web and AI applications with Flutter, Firebase, React, Python and the MERN stack. Seeking a software engineering internship to gain industry experience, learn from experienced professionals and contribute to innovative projects.

EDUCATION

BSc (Hons) in Software Engineering — **2022 – 2026**
University of Plymouth (UK) · NSBM Green University
Second Class Honours (Upper Division)

Foundation Programme in Computing — **2022 – 2023**
NSBM Green University

Secondary Education
Nugegoda Sujatha Vidyalaya

TECHNICAL SKILLS

- Mobile App Development
- Web Development
- Artificial Intelligence (AI)
- UI/UX Design
- Flutter
- Firebase
- React.js
- HTML, CSS, JavaScript
- Java
- C#
- Python

PROJECTS

MediNext – Smart Pharmacy Mobile App | Final Year Project, 2026 — Flutter · Firebase · OSRM API

- Developed an Android mobile application using Flutter and Firebase for real-time data synchronization and secure role-based authentication.
- Integrated the OSRM API for geolocation-based pharmacy mapping and navigation, alongside an AI virtual health assistant and automated medication reminders.
- Built a pharmacist dashboard to manage live inventory and digital prescriptions, with bidirectional real-time chat for patient–pharmacist communication.

The Book Parlour – Semantic Book Recommender — Python · Flask · ChromaDB · LangChain

- Built an AI-powered book recommendation system using vector embeddings and k-NN similarity search via ChromaDB.
- Applied sentiment analysis and text classification to build multi-axis emotional profiles for precision filtering.
- Developed a REST API backend with a custom web UI for seamless, dynamic book discovery.

NexCart – Full-Stack E-Commerce Platform — MERN · Socket.io · Docker · JWT

- Developed a Docker-containerized e-commerce app using the MERN stack.
- Implemented real-time inventory and order-tracking updates using Socket.io.
- Configured secure JWT authentication and role-based access control for customer and admin interfaces.

Archify – 3D Interior Design & Visualization — JavaScript · Firebase · Three.js

- Created a web application featuring an interactive 2D floor planner using JavaScript and Firebase.
- Rendered real-time 3D room visualizations with dynamic lighting and shadows using Three.js.

Gait-Based Verification — AI Biometrics · Neural Networks

- Engineered an AI biometric authentication system using a Feed-Forward Neural Network (FF-MLP).
- Processed sensor data with mRMR feature selection, achieving 84.14% identification accuracy.

REFERENCES

Diluka Wijesinghe
Lecturer, Software Engineering & Computer Security · NSBM Green University
Email: Diluka.w@nsbm.ac.lk

D. M. Kaushalya Abeyssinghe
Attorney at Law · LLM, Cardiff University (UK)
Phone: +94 72 447 3856 · **Email:** dmkabey999@gmail.com